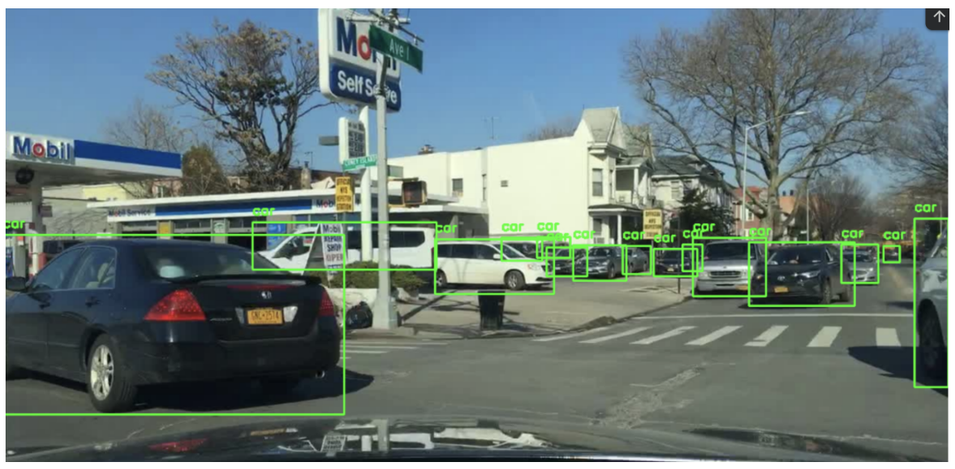
Day vs. Night Object Detection: YOLOv8 on BDD100K – Deep Machine Learning

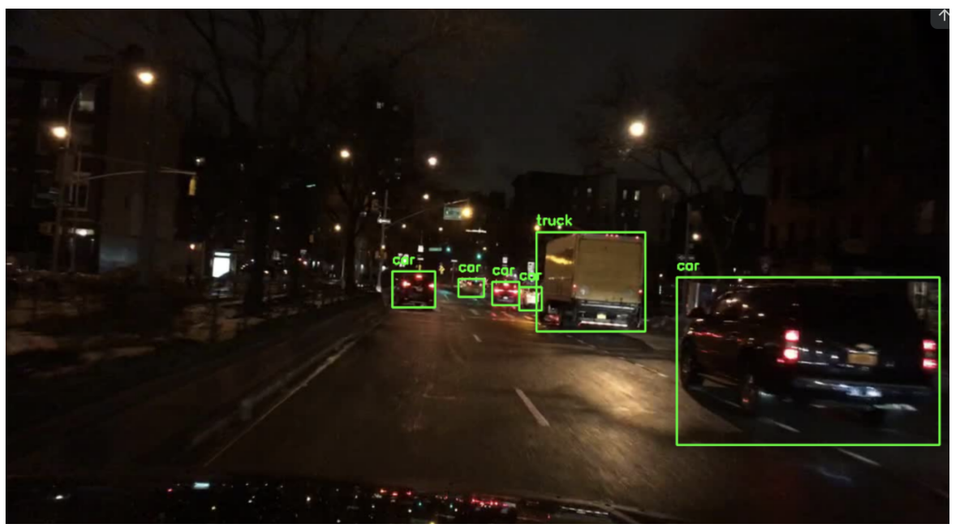
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Introduction

Reliable object detection is essential for autonomous driving. Reduced illumination, glare, and low contrast can make nighttime detection challenging. We study how the **lighting conditions used during training** affect YOLOv8n detection performance.





Study objectives

- Compare detection quality on day and night images.
- Test balanced, night-only, and day-only training.
- Analyze accuracy and the day–night performance gap.

Theory and Evaluation

YOLOv8n is a lightweight, single-stage object detector. It predicts object locations and categories in one forward pass. **Transfer learning** adapts COCO-pretrained weights to five BDD100K classes: person, bicycle, car, bus, and truck.

- **Precision:** proportion of predicted detections that are correct.
- **Recall:** proportion of ground-truth objects detected.
- **mAP50:** mean average precision at IoU $= 0.50$.
- **mAP50–95:** mean AP across IoU thresholds 0.50 to 0.95.

$$\mathrm{IoU} = \frac{|B_{\mathrm{pred}} \cap B_{\mathrm{gt}}|}{|B_{\mathrm{pred}} \cup B_{\mathrm{gt}}|}, \qquad \mathrm{Precision} = \frac{TP}{TP + FP}.$$

Experimental dataset: 10,000 selected BDD100K images, equally split between daytime and nighttime. Official training images supply the training set; the official validation source is partitioned into disjoint validation and held-out test subsets.

Split	Day	Night	Total
Train	3,500	3,500	7,000
Validation	750	750	1,500
Held-out test	750	750	1,500

Training settings: 20 epochs, 640-pixel input, batch size 8, seed 42. All fine-tuned models start from the same COCO-pretrained YOLOv8n weights.

Experimental Design

Experiment	Training data	Images
E0: Pretrained	COCO weights; no BDD100K fine-tuning	0
E1: Balanced	3,500 day + 3,500 night	7,000
E2: Night-only	Nighttime images only	7,000
E4: Day-only	Daytime images only	7,000

Evaluation: The same 750 daytime and 750 nighttime validation images are used for E1, E2, and E4. E0 is retained as a pretrained baseline but excluded from the numerical comparison until its COCO class IDs are consistently mapped to the five BDD100K classes.

Results: Day vs. Night

Validation mAP50 (%): Higher is better. The table contains measured results, not illustrative values.

Model	Day	Night	Day − Night
E1 Balanced	**45.23**	**47.91**	−2.67 pp
E2 Night-only	32.15	39.48	−7.33 pp
E4 Day-only	44.74	37.43	+7.31 pp

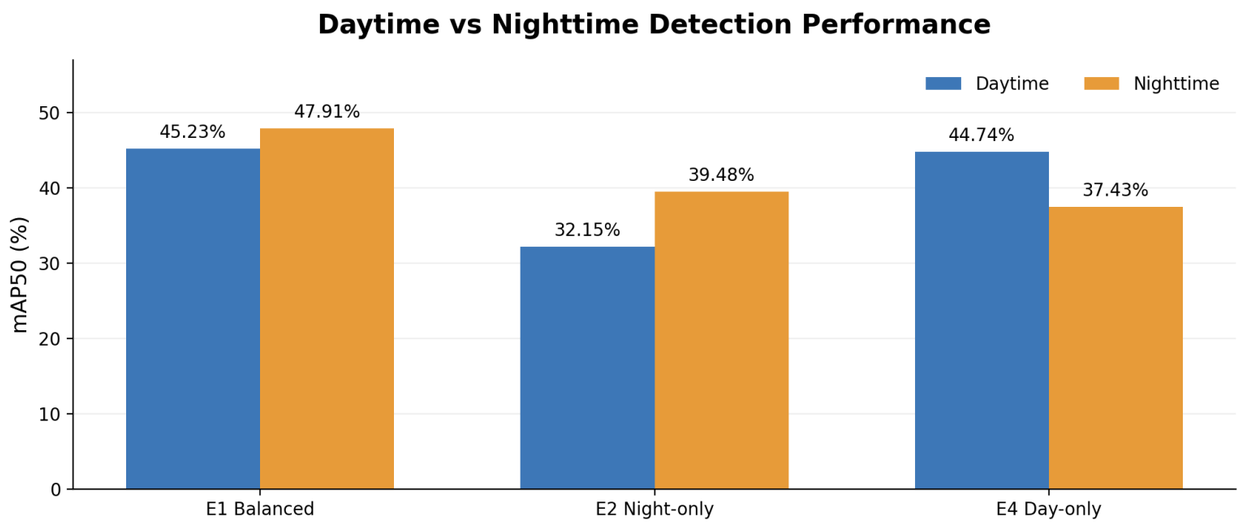


Figure 1: Daytime and nighttime mAP50 for E1, E2, and E4. Supply the chart exported from Colab.

Validation mAP50–95 (%):

Model	Day	Night
E1 Balanced	27.04	**28.34**
E2 Night-only	18.74	22.76
E4 Day-only	**27.14**	21.57

Main observation: E1 exceeds E2 by **8.43 percentage points** in nighttime mAP50 and E4 by **10.48 points**. E4 remains competitive on daytime data.

Discussion and Conclusion

- **Balanced training performs best at night** and maintains strong daytime detection.
- **Condition-specific training favors its own lighting domain:** E2 has higher night than day mAP50; E4 shows the reverse.
- **The smaller E1 gap** (2.67 points in absolute value) suggests more consistent validation performance across lighting conditions.
- Day/night validation sets also differ in object composition; reported values are **validation**, not held-out test, results.

Conclusion: Among the three fine-tuned YOLOv8n models, balanced training provides the strongest overall cross-condition performance, particularly at night. Further controlled training-budget comparisons and held-out test evaluation are needed to confirm generalization.

References

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